

001

002

003

Exploring RGB and Video V-JEPA 2.1 Features
for Aerial Depth Estimation and Semantic Scene
Completion

001

002

003

004

Anonymous ACCV 2026 Submission

004

005

Paper ID #1107

005

006

007

008

009

010

011

012

013

014

015

016

017

018

019

020

021

022

023

024

025

026

027

028

Abstract. Aerial 3D scene understanding is essential for autonomous UAV operations such as cargo delivery, search-and-rescue, and human-UAV collaboration. Camera-based perception is attractive for these settings given its light weight, low cost, availability, and compatibility with UAV size, weight, and power constraints. This motivates methods that can infer geometry, semantics, and free space from visual observations, making Monocular Depth Estimation (MDE) and Semantic Scene Completion (SSC) key components of UAV 3D perception that enable 3D scene understanding. Aerial imagery introduces altitude-dependent scale variation, oblique viewpoints, weak texture, and different occlusion patterns, demanding domain-specific solutions. Using OccuFly, a real-world aerial dataset with RGB images, metric depth maps, and semantic occupancy grids, we study whether V-JEPA 2.1 foundation features transfer to aerial MDE and SSC. Our ablations progress from V-JEPA 2.1 image-spatial & video-spatiotemporal feature handling, from linear probing to DPT (Dense Prediction Transformer)-style multiscale decoding, FiLM (Feature-wise Linear Modulation)-based altitude & intrinsics conditioning for MDE, followed by injecting conditioned depth-estimation priors, deploying a learnable UAV pose-aware multi-view stereo depth-estimation module, and transferring spatiotemporal features for SSC. Through this staged analysis, we separate the contributions of representation transfer, altitude conditioning, and depth-estimation variants to improving aerial SSC.

006

007

008

009

010

011

012

013

014

015

016

017

018

019

020

021

022

023

024

025

026

027

028

029

030

031

032

Keywords: Aerial · 3D Scene Understanding · UAV Perception · Monocular Depth Estimation · Semantic Scene Completion · Semantic Occupancy Prediction · V-JEPA 2.1 · Video Foundation models · 2D-3D lifting · Pose-aware multi-view stereo depth · OccuFly

029

030

031

032

033

1 Introduction

033

034

035

036

037

038

Aerial 3D scene understanding is essential for autonomous UAV operations such as cargo delivery, search-and-rescue, and human-UAV collaboration. In these settings, an autonomous platform must not only recognize visible image regions, but also infer metric geometry, free space, and the semantic occupancy of partially observed volumes. Such a representation is important for collision avoidance,

034

035

036

037

038

viewpoint planning, landing-zone assessment, and interaction with objects and humans in cluttered outdoor environments. Camera-based perception is particularly attractive for UAVs because RGB cameras are lightweight, inexpensive, widely available, and compatible with strict size, weight, and power constraints. However, inferring dense 3D structure from monocular or short video observations remains substantially harder than recognition, since the image formation process collapses scale, depth, and occlusion into a two-dimensional signal. Aerial video offers an additional source of geometry through motion parallax, but exploiting it requires more than generic temporal feature aggregation: the system must relate target and source views through camera calibration and relative pose.

Most progress in semantic scene completion (SSC) and camera-based 3D perception has been driven by indoor reconstruction and autonomous-driving benchmarks. Early SSC formulations used depth as input to jointly predict volumetric occupancy and semantics [30]. Later monocular methods, such as MonoScene [5], moved the problem toward RGB-only 2D-to-3D lifting, while VoxFormer [15], TPVFormer [11], SurroundOcc [32], OpenOccupancy [31], CGFormer [37], and FoundationSSC [6] developed increasingly capable view-transformation, voxel-attention, and semantic-geometric fusion designs. These works demonstrate that explicit 3D representations are valuable for scene-level reasoning, but their primary evaluation domains remain road scenes with ground-level perspective, relatively constrained camera height, and characteristic automotive object layouts. Aerial viewpoints violate many of these assumptions: altitude changes the apparent scale of all objects, oblique views produce mixed top-down and side-view evidence, fine semantic boundaries can be weakly textured, and occlusions differ from those observed by ground vehicles. Consequently, strong methods trained or designed for terrestrial scenarios do not necessarily transfer to UAV perception without careful adaptation.

OccuFly [8] creates an opportunity to study this gap in a real-world aerial setting. It provides RGB images, metric depth maps, camera calibration, and semantic occupancy grids for aerial scenes captured at multiple altitudes and across urban, industrial, and rural environments. Unlike earlier UAV semantic segmentation datasets such as UAVid [20], which focus on 2D dense labeling, or aerial reconstruction resources such as UrbanScene3D [18], OccuFly directly supports the joint study of metric depth estimation and 3D semantic scene completion from aerial camera observations. This makes it possible to ask whether modern visual foundation representations provide transferable geometric and semantic priors for aerial 3D scene understanding, rather than only for image classification, 2D segmentation, or ground-level depth prediction. It also enables a more precise question: when multiple calibrated UAV views are available, does explicit multiview correspondence provide a stronger geometric prior for SSC than a monocular depth estimator alone?

In this work we investigate V-JEPA 2.1 [23] as a frozen foundation representation for aerial depth estimation and depth-aware semantic scene completion. V-JEPA and its successors learn by predicting representations in latent space

rather than reconstructing pixels [2, 3]. V-JEPA 2.1 further emphasizes dense predictive supervision, deep self-supervision at intermediate layers, and unified image/video tokenization, making it a natural candidate for dense 3D perception. The central question of this paper is not whether a large foundation model performs well on generic vision tasks, but whether its spatial and spatio-temporal features encode the aerial geometry needed for metric depth and volumetric semantic completion. We therefore keep the backbone frozen and vary the decoder, metadata conditioning, temporal aggregation, and geometric lifting mechanism.

We study this question along three axes. First, for monocular depth estimation, we compare frozen linear probes against structured DPT-style decoders [27]. The linear probes test how linearly accessible metric depth is from V-JEPA 2.1 features, while the DPT decoders test whether multiscale token reassembly and convolutional fusion can recover fine spatial detail from frozen transformer features. We also examine a Simple Feature Pyramid (SFP) adapter inspired by ViTDet [14], which converts a single final-layer feature map into a pyramid suitable for dense prediction. This allows us to study the dense final-layer representation produced by V-JEPA 2.1, which, according to its authors, incorporates information from earlier layers into the final feature map. Because aerial scale is strongly coupled to flight altitude and camera calibration, we further condition selected decoders with feature-wise linear modulation (FiLM) [25] using altitude and camera intrinsics. Second, we compare spatial RGB features with spatio-temporal video features. Aerial videos contain motion parallax and temporal continuity that may help resolve depth ambiguities.

Third, for SSC, we compare monocular depth priors inside depth-aware lifting architectures inspired by Lift-Splat-Shoot [26], CGFormer [37], and FoundationSSC [6]. The monocular V-JEPA/FoundationSSC branch uses a DPT-FiLM depth-probability frontend to lift semantic context into the OccuFly voxel grid. We further introduce a Pose-MVS-Cost-Volume FoundationSSC variant that replaces the monocular depth-probability frontend with a posed multiview stereo cost volume in the spirit of learned MVS cost-volume methods [35]. Given a target UAV view and neighboring calibrated source views, this branch matches target/source V-JEPA video features across metric depth hypotheses, regularizes the resulting cost volume, and converts it into the depth distribution consumed by the FoundationSSC lifting module. We also evaluate a Pose-MVS-FiLM variant that applies altitude/intrinsics FiLM conditioning directly to the MVS depth-bin logits before the depth softmax, so that the dense depth map and lifting probabilities remain consistent.

This design separates two different uses of video. Dense temporal attention tests whether frozen V-JEPA video tokens contain useful target-frame depth, semantic context and world reasoning cues. In contrast, Pose-MVS tests an explicitly geometric hypothesis: calibrated source views should constrain depth through projective correspondence, even when monocular appearance cues are ambiguous. This distinction is important for aerial perception, where altitude variation, repeated textures, vegetation, roofs, and oblique viewing directions

can make single-image depth uncertain while neighboring UAV frames may provide exploitable parallax.

Our contributions are as follows:

- We present a systematic study of frozen V-JEPA 2.1 spatial and spatio-temporal representations for aerial metric depth estimation on OccuFly.
- We compare representation probes of increasing structure: linear depth heads, DPT-style multiscale decoders, SFP-adapted final-layer decoders, and altitude/intensity-conditioned FiLM variants.
- We analyze the role of spatio-temporal features through dense temporal-attention adapter.
- We extend the investigation from depth to semantic scene completion by incorporating predicted depth and depth distributions as geometric priors in CGFormer-, and FoundationSSC-inspired aerial SSC variants.
- We introduce Pose-MVS-Cost-Volume FoundationSSC, which replaces monocular depth-probability lifting with pose-aware multiview cost-volume depth estimation from target/source UAV views.
- We further evaluate Pose-MVS-FiLM, a metadata-conditioned MVS variant that modulates cost-volume depth logits with altitude and camera intrinsics while preserving probability depth consistency for voxel lifting.
- We identify the practical domain gap between ground-level camera-centric 3D perception and aerial 3D scene understanding, providing a focused evaluation protocol for future foundation-model transfer studies in UAV perception.

2 Related Work

2.1 Aerial scene understanding and aerial 3D benchmarks

Aerial perception has traditionally been studied through 2D semantic segmentation, object detection, mapping, and reconstruction. UAVid [20] introduced high-resolution oblique UAV videos for semantic segmentation and highlighted challenges such as scale variation, moving objects, and temporal consistency. Other aerial resources, including UrbanScene3D [18], support large-scale urban reconstruction and simulation from aerial observations. These datasets are valuable for 2D and reconstruction-oriented aerial perception, but they do not directly provide the combination of RGB images, metric depth, camera calibration, and per-frame semantic occupancy grids needed to study aerial SSC.

OccuFly [8] addresses this limitation by introducing a real-world aerial benchmark for camera-based 3D semantic scene completion. It provides aerial RGB images, metric depth maps, semantic voxel grids, and camera metadata over multiple scenes, altitudes, environments, and seasons. The dataset is especially relevant for UAV perception because it captures the geometric ambiguity of low-altitude flight: object scale depends on altitude, many surfaces are seen from oblique or partially top-down viewpoints, and free-space reasoning must be performed in a 3D volume rather than only on the image plane. Our work

uses OccuFly not merely as an evaluation dataset, but as a testbed for studying whether dense image/video foundation features and explicit multiview geometry transfer to aerial depth and semantic occupancy.

2.2 Monocular depth estimation and dense prediction decoders

Monocular depth estimation is inherently ambiguous because metric depth must be inferred from perspective, texture, object size, scene layout, and learned priors. Early deep approaches used coarse-to-fine convolutional networks and scale-aware objectives [7]. Later methods improved dense prediction through stronger encoders, multiscale decoders, and more diverse training data. MiDaS demonstrated that mixing heterogeneous datasets with compatible objectives improves cross-dataset robustness [28]. DPT replaced purely convolutional backbones with vision transformers and reassembled intermediate tokens into multiscale image-like features for dense prediction [27]. ZoeDepth combined relative-depth generalization with metric-depth specialization [4], while Depth Anything and Depth Anything V2 scaled monocular depth training with large pseudo-labeled and synthetic data, producing robust general-purpose depth models [33, 34].

These models show that large-scale pretraining and dense decoders are crucial for depth, but aerial depth remains a distinct transfer problem. UAV images often lack the ground-level cues used by indoor and automotive models: the horizon may be absent, object size priors vary with altitude, repetitive roofs and vegetation create weak texture, and far-range depth errors can dominate voxel lifting. For this reason we compare lightweight linear probe, DPT-style decoders on frozen V-JEPA 2.1 features, evaluate final-layer SFP adaptation inspired by ViTDet [14], and condition selected decoders with altitude and camera intrinsics through FiLM [25]. This lets us separate three factors: the geometric information already present in the representation, the spatial reconstruction capacity of the decoder, and the usefulness of explicit aerial metadata.

2.3 Multiview stereo and cost-volume geometry

Monocular depth relies heavily on learned priors, whereas multiview stereo (MVS) estimates depth by matching observations across calibrated views. Learning-based MVS methods such as MVSNet build a plane-sweep cost volume by warping source-view features into a reference view over multiple depth hypotheses, regularize the cost volume with 3D convolutions, and infer depth from a probability distribution over planes [35]. Recurrent MVSNet reduces the memory footprint of high-resolution MVS by replacing full 3D cost-volume regularization with recurrent processing over depth hypotheses [36]. These methods are designed for depth reconstruction, but the underlying principle is directly relevant to aerial SSC: when camera poses are available, target/source correspondence can provide geometric evidence that is complementary to monocular appearance priors.

Our Pose-MVS variants adapt this cost-volume principle to frozen V-JEPA 2.1 video features and downstream semantic scene completion. Rather than producing only a target-view depth map, the MVS branch predicts a categorical depth

distribution aligned with the FoundationSSC lifting bins. The resulting distribution is used to place semantic context into the OccuFly voxel grid, making MVS a geometric frontend for SSC.

In parallel, our Dense Temporal Attention adapter is inspired by the attentive probe mechanism used in the official V-JEPA 2 evaluation pipeline, where frozen video tokens are aggregated by lightweight attention-based heads for downstream classification tasks [2, 23]. In the released V-JEPA 2 repository, this design appears in the classification example as an attentive classifier that pools patch-wise video features using learnable query tokens before applying a task-specific prediction head. We adapt this idea from global action recognition to dense geometric prediction: instead of collapsing the video into a single classification embedding, the adapter learns to aggregate temporal V-JEPA tokens into depth-aware context features that can support downstream lifting and semantic scene completion.

2.4 Self-supervised visual foundation models

Self-supervised visual foundation models aim to learn transferable representations without task-specific labels. Masked autoencoders showed that high-ratio masked image reconstruction can scale vision transformers effectively [9]. I-JEPA introduced a non-generative joint-embedding predictive architecture for images, predicting target-block representations from context representations instead of reconstructing pixels [1]. DINOv2 demonstrated that carefully curated large-scale self-supervised training can yield robust all-purpose image features [24], and DINOv3 further emphasizes high-resolution dense features and broad transfer across natural and aerial imagery [29].

Video representation learning extends these ideas to temporal structure. V-JEPA learns video representations by predicting masked spatio-temporal features in latent space and can be used as a frozen backbone with lightweight probes [3]. V-JEPA 2 scales this approach to internet-scale video and image data for understanding, prediction, and planning [2]. V-JEPA 2.1 further modifies the recipe toward dense image/video features through dense predictive loss, deep self-supervision, and multimodal tokenizers [23]. These properties make V-JEPA 2.1 attractive for aerial 3D perception, where one needs spatial layout, temporal consistency, and view-dependent geometry. However, strong generic dense features do not guarantee metric depth accuracy or reliable 3D lifting under aerial viewpoints. Our work therefore evaluates V-JEPA 2.1 through explicit downstream depth, cost-volume, and SSC probes rather than only through representation visualization or global recognition metrics.

2.5 2D-to-3D lifting and camera-based semantic scene completion

Semantic scene completion jointly estimates occupied and free voxels together with semantic labels. SSCNet introduced the task using single-view depth input and 3D convolutional reasoning [30]. MonoScene extended SSC to monocular RGB by projecting 2D features into 3D and using 3D context priors [5].

Camera-based outdoor SSC then evolved toward sparse voxel queries, deformable attention, and explicit view transformation. VoxFormer uses estimated depth to initialize sparse visible voxel queries and then densifies them with transformer reasoning [15]. TPVFormer represents 3D space through three orthogonal planes, improving over a single BEV plane for semantic occupancy prediction [11]. SurroundOcc and OpenOccupancy expanded the problem toward surrounding multi-camera occupancy perception in autonomous driving [31, 32].

A recurring theme in these works is the importance of depth-aware lifting. Lift-Splat-Shoot first formalized a camera-to-BEV representation by predicting a depth distribution for each image feature and splatting frustum features into a common spatial grid [26]. BEVDepth later showed that explicit depth supervision improves camera-based BEV 3D detection [16]. For SSC, CGFormer addresses depth ambiguity by generating context-dependent voxel queries and extending deformable cross-attention from 2D image space to 3D pixel space; it also combines voxel and tri-perspective-view representations to capture local and global 3D context [37]. FoundationSSC argues that semantic and geometric priors should be decoupled at both source and pathway levels, using foundation features and geometric depth distributions before hybrid 3D fusion [6].

Our SSC design follows this line of work but differs in domain and experimental goal. Existing methods are primarily optimized for indoor or autonomous-driving scenes, whereas we target aerial observations from OccuFly. We use predicted monocular depth and depth probabilities as geometric priors for voxel lifting, while V-JEPA 2.1 features provide semantic context. We then add Pose-MVS-Cost-Volume FoundationSSC to test whether calibrated target/source matching improves the depth distribution used for lifting, and Pose-MVS-FiLM to test whether altitude and intrinsics should condition that multiview distribution. This allows us to compare three sources of geometry under a shared aerial SSC protocol: monocular depth priors, video-feature temporal aggregation, and explicit posed multiview correspondence.

2.6 Position of this work

The closest work to ours is OccuFly [8], which establishes the aerial SSC benchmark and reports baseline limitations of existing methods. Our work is complementary: we use OccuFly to analyze the transfer behavior of a recent video/image foundation model for coupled aerial 3D tasks. Compared with generic monocular depth models such as Depth Anything V2 [34], we focus on what can be recovered from frozen V-JEPA 2.1 features and how decoder structure, meta-data conditioning, temporal context, and posed multiview matching affect metric depth. Compared with standard MVS methods such as MVSNet and R-MVSNet [35, 36], our goal is not standalone depth reconstruction but semantic occupancy: the cost volume is used to produce the depth-probability tensor that drives FoundationSSC-style voxel lifting. The resulting study connects aerial benchmarks, dense foundation representations, multiview cost-volume geometry, and depth-aware semantic scene completion.

3 Method

This section describes our evaluation framework for transferring frozen V-JEPA 2.1 image-spatial and video-spatiotemporal features to aerial metric depth estimation and semantic scene completion (SSC). The method starting is intentionally organized as a sequence of probes followed by progressively more structured decoders for evaluating dense feature transfer in both rgb and video variants of V-JEPA 2.1 for aerial MDE, with aim to select the best performer and compare with Occufly dataset Finetuned DAV2 MDE. Decoupled Geometric and Image context layers style SSC paradigm is chosen, specifically CGFormer and FoundationSSC inspired variants as they structurally allow for controlled evaluation on geometric and image context feature contribution for SSC.

We evaluate if and how far previously selected learned MDE heads, a learnable pose aware Multi View Stereo Depth Estimation head and a learnable image context feature represenattion head with their backbone as frozen V-JEPA 2.1, benifit in improving aerial SSC. Knowing that existing metric MDE models are not generalisable we leverage the fact that UAV pose/altitude and onboard camera intrinsics are usually known or estimated using auxilary systems and evaluate benifits of Feature wise learning based modulation (FiLM) for aerial MDE and indirectly SSC.

We mathematically formulate the train/eval/test data source, problem statement for MDE and SSC. Followed by description of the adapter/neck needed to make V-JEPA 2.1 features usable by the downstream heads. Then progressively dive deeper into considered MDE heads and SSC variants related sourcing, compatability details followed by answering a questionair that answers what componet or componet combination has helped to benifit aerial SSC?? This direction of flow is summarized in Fig. 1. The following sections outline the evaluation framework, metrics, and experiments conducted. Later the same structure will be used to report results obtained.

3.1 Problem Formulation and Notation

We use Occufly dataset [8] for all experiments. The per sample data details used from the dataset is mathematically formulated as follows.

$$s_t = (I_t, D_t, M_t, Y_t, K_t, h_t, T_t^{w \rightarrow c}), \quad (1)$$

where $I_t \in \mathbb{R}^{3 \times H \times W}$ is an RGB frame, D_t is a metric depth map, $M_t \in \{0, 1\}^{H \times W}$ is the valid-depth mask, Y_t is the semantic occupancy grid, K_t is the camera intrinsic matrix, h_t is the UAV altitude, and $T_t^{w \rightarrow c} \in \mathbb{R}^{4 \times 4}$ is the world-to-camera pose. The leading dimension 3 in I_t denotes RGB color channels, $H \times W$ denotes the image resolution before the V-JEPA 2.1 resize, and the 4×4 pose matrix is a homogeneous rigid transform.

RGB		V-JEPA 2.1 ViTG		Video			
				Dense Temporal Attention (DTA)			
MDE - Depth Heads		SSC		MDE - Depth Heads		SSC	
1. Linear Probe RGB	CGFormer variant		2. Linear Probe Video	6.FSSC_video_context DTA_DPT_depth_DTA_FILM_DPT			
	Image Context: Direct			Image Context: DTA_DPT			
3. Simple Feature Pyramid - Dense Prediction Head (SFP_FILM_DPT)	Geometry:	Depth priors		Geometry: DTA_FILM_DPT			
		1. DAV2 Finetuned Occufly					
		2. FILM_DPT		4.FSSC_video_context DTA_DPT_depth_DTA_mvs			
	3.FSSC_RGB_context DPT_depth_Film_DPT			Image Context: DTA_DPT			
4. Multi Layer - Dense Prediction Head (DPT)	Image Context: DPT		6. DTA_FILM_DPT_Video	Geometry: DTA_Pose_mvs			
5. Multi Layer - Featurewise Linear Modulation - Dense Prediction Head (FILM_DPT_RGB)	Geometry: FILM_DPT						
	4.FSSC_RGB_context FILM_DPT_depth_FILM_DPT			5.FSSC_video_context DTA_DPT_depth_DTA_FILM_mvs			
	Image Context: FILM_DPT			Image Context:DTA_FILM_DPT			
	Geometry: FILM_DPT			Geometry: FILM_Pose_MVS			

Fig. 1: Evaluation overview for the dense feature-transfer study. The figure summarizes the progression from frozen V-JEPA 2.1 RGB/video representations to monocular and multiview depth estimation heads, followed by the SSC variants that use the selected context and geometry pathways.

The monocular depth, multiview depth, and SSC task functions are

$$\begin{aligned}
 f_D^{\theta_D} &: (I_t, K_t, h_t) \mapsto \hat{D}_t, \\
 f_{D,MVS}^{\theta_M} &: (\mathcal{V}_t, h_t) \mapsto (\hat{D}_t^{MVS}, P_t^{D,MVS}), \\
 f_S^{\theta_S} &: (\Phi_t, K_t, h_t, \hat{D}_t, P_t^D) \mapsto \hat{Y}_t.
 \end{aligned} \tag{2}$$

Here $f_D^{\theta_D}$ denotes the monocular depth-estimation model with trainable depth parameters θ_D , and $f_{D,MVS}^{\theta_M}$ denotes the learnable pose-aware multiview stereo depth estimator with parameters θ_M . The multiview input is

$$\mathcal{V}_t = \{(I_{t_i}, K_{t_i}, T_{t_i}^{w \rightarrow c})\}_{i=0}^{N_s}, \quad t_0 = t, \tag{3}$$

Here t is the target time index for which depth and semantic occupancy are predicted. The notation $t_0 = t$ means that the first element of $\mathcal{V}_t$ is the target, or reference, view. The remaining elements $t_1, \dots, t_{N_s}$ are calibrated source views used for multiview matching against the target view. We consider only two views for this study.

In the implementation, $f_{D,MVS}^{\theta_M}$ is realized by source-specific V-JEPA 2.1 video features, pose-aware projective matching, and a trainable cost-volume regularizer. At this level, we write it generally as a mapping from posed views to a target-frame dense depth estimate $\hat{D}_t^{MVS}$ and depth probability tensor $P_t^{D,MVS}$.

The function $f_S^{\theta_S}$ denotes the semantic scene completion model with trainable SSC parameters θ_S . In our frozen-backbone setting, the V-JEPA 2.1 encoder weights are fixed, while θ_D , θ_M , and θ_S are either learnable or frozen based on the variant. The term Φ_t denotes the visual context representation extracted from the target observation. The camera intrinsics K_t and altitude h_t provide geometric metadata for projection and scale conditioning. The dense depth prediction $\hat{D}_t$ gives the expected metric depth at each image location, while P_t^D is a depth probability tensor used for lifting image features along camera rays. The output $\hat{Y}_t$ is the predicted semantic occupancy grid.

3.2 Frozen V-JEPA 2.1 Feature Extraction

V-JEPA 2.1 is used in two input modes: an RGB/image mode and a video mode. Let E_θ denote the frozen V-JEPA 2.1 encoder [23]. For the ViT-Gigantic checkpoint used in the experiments, the feature channel dimension is $C = 1664$, which is why every spatial token is represented by a 1664-dimensional descriptor. The model has 48 transformer blocks, indexed from 0 to 47, and a 384×384 input with patch size 16 producing a 24×24 spatial grid, $384/16 = 24$.

RGB mode: A target image I_t is encoded into one or more spatial feature maps. We use either the final layer alone or four intermediate layers:

$$\mathcal{L}_G = \{11, 23, 37, 47\}, \quad F_t^{(l)} = E_\theta^{(l)}(I_t) \in \mathbb{R}^{C \times 24 \times 24}, \quad l \in \mathcal{L}_G. \quad (4)$$

Here, $\mathcal{L}_G$ lists the selected ViT-G layers, l indexes one of those layers, and $F_t^{(l)}(:, u, v)$ is the 1664-dimensional RGB descriptor at spatial grid cell (u, v) . For image-mode depth prediction, the frozen encoder returns image-grid features without a temporal tubelet axis. In particular, the final-layer RGB feature map has the form

$$F_t^{(47)} \in \mathbb{R}^{1664 \times 24 \times 24}. \quad (5)$$

A dense linear probe applies a lightweight batch-normalized 1×1 prediction head to the final frozen feature map, following standard frozen-feature linear probing practice for dense ViT representations [21, 23, 29]. The DPT variant instead consumes selected intermediate feature maps and reassembles them into a multiscale hierarchy following DPT [27].

Video mode: We form a four-frame clip centered at target frame t :

$$X_t = [I_{t+\tau_1}, I_{t+\tau_2}, I_{t+\tau_3}, I_{t+\tau_4}] \in \mathbb{R}^{3 \times 4 \times 384 \times 384}, \quad (6)$$

$$[\tau_1, \tau_2, \tau_3, \tau_4] = [-1, 0, 0, +1].$$

The duplicated zero offset places the target frame twice in the clip. This gives a target-centered local temporal window while keeping the clip length divisible by the V-JEPA 2.1 tubelet size.

For V-JEPA 2.1 ViT-Gigantic, the patch size is 16, the video tubelet size is 2, and the feature dimension is $C = 1664$. Therefore, a 384×384 input produces

a 24×24 spatial grid, and a T -frame video clip produces $T' = T/2$ temporal tubelet positions:

$$G_t^{(l)} = E_\theta^{(l)}(X_t) \in \mathbb{R}^{1664 \times T' \times 24 \times 24}. \quad (7)$$

For the four-frame clips used here, $T = 4$ and hence $T' = 2$. Video-mode features contain both spatial locations and temporal tubelet tokens, so we use lightweight adapters to expose them to dense depth heads.

We tested two types of adapters for video features: The first type being *linear probe for video mode* that uses only the final V-JEPA feature tensor $G_t^{(47)}$. To keep the probe linear while retaining the tubelet information, the temporal tubelet axis is folded into the channel axis before applying the same 1×1 depth-bin head used by the image linear probe:

$$\begin{aligned} G_t^{(47)} \in \mathbb{R}^{1664 \times T' \times 24 \times 24} &\mapsto \tilde{F}_t \in \mathbb{R}^{(1664T') \times 24 \times 24}, \\ \tilde{F}_t[t'C + c, u, v] &= G_t^{(47)}[c, t', u, v], \quad 0 \leq c < 1664, \quad 0 \leq t' < T'. \end{aligned} \quad (8)$$

For $T = 4$, this gives

$$\tilde{F}_t \in \mathbb{R}^{3328 \times 24 \times 24}. \quad (9)$$

This probe has no spatial decoder, temporal attention, recurrent module, or backbone finetuning. It is used as a diagnostic for whether depth-relevant temporal information is linearly accessible from the frozen final-layer video features.

The second adapter is the *Dense Temporal Attention* (DTA) adapter. DTA is inspired by the attentive probing heads used for V-JEPA 2.1 video evaluation, where learnable-query cross-attention pools frozen video tokens for classification [2, 22, 23]. Instead of pooling all spatio-temporal tokens into one global classification vector, DTA applies the same learned-query attention idea locally at every spatial cell. For each selected layer $l \in \{11, 23, 37, 47\}$, the temporal descriptors at location (u, v) are

$$S_{t,u,v}^{(l)} = \left[G_t^{(l)}[:, 1, u, v], \dots, G_t^{(l)}[:, T', u, v] \right] \in \mathbb{R}^{T' \times 1664}. \quad (10)$$

The adapter A_l attends over this local tubelet sequence and returns one image-grid feature vector at the same spatial location:

$$A_l(G_t^{(l)}) = \bar{F}_t^{(l)} \in \mathbb{R}^{1664 \times 24 \times 24}. \quad (11)$$

At a single spatial location, the operation can be summarized as

$$\begin{aligned} \mu_t^{(l)}[:, u, v] &= \frac{1}{T'} \sum_{\tau=1}^{T'} G_t^{(l)}[:, \tau, u, v], \\ \bar{F}_t^{(l)}[:, u, v] &= \mu_t^{(l)}[:, u, v] + \alpha_l \left(\text{DTA}_l(S_{t,u,v}^{(l)}) - \mu_t^{(l)}[:, u, v] \right), \end{aligned} \quad (12)$$

where $\text{DTA}_l(\cdot)$ denotes the bottleneck projection, learned-query multi-head temporal attention, MLP refinement, and expansion back to 1664 channels. The

Input split	OccuFly train/validation/test samples
↓	image (I_t, K_t, h_t) , video clip (X_t, K_t, h_t) , or posed views $\mathcal{V}_t$
Frozen backbone	V-JEPA 2.1 features $F_t^{(l)}$ or $G_t^{(l)}$
↓	LP / SFP-DPT / DPT-FiLM / Video-FiLM-DPT / Pose-MVS-FiLM
Trainable head	depth-bin logits
↓	
Output	depth probabilities P_t^D and estimated depth $\hat{D}_t$ or MVS counterparts $(P_t^{D,MVS}, \hat{D}_t^{MVS})$

Fig. 2: Common depth-estimation flow. The V-JEPA 2.1 backbone is frozen, while the depth head is trained on the official OccuFly train/validation/test split.

residual gate α_l is initialized near temporal mean pooling, so training can gradually introduce attention-based temporal corrections.

The resulting DTA maps $\{\bar{F}_t^{(l)}\}_{l \in \{11, 23, 37, 47\}}$ have the same $1664 \times 24 \times 24$ image-grid form as RGB V-JEPA feature maps. They can therefore be passed to the same dense downstream machinery used in image mode, including DPT-style decoding for depth prediction, F2D feature prediction, and pose aware multiview cost-volume construction.

3.3 Depth Estimation

Depth estimation in this study follows the monocular mapping $f_D^{\theta_D}$ and the pose-aware multiview stereo mapping $f_{D,MVS}^{\theta_M}$ in Eq. (2). The monocular heads use a target image, camera intrinsics, and altitude metadata, while the MVS heads use calibrated target/source views to estimate target-frame depth. In the variants considered here, each depth head predicts ray-wise logits over metric depth bins and converts them to a dense metric depth estimate:

$$p_{k,u,v} = \text{softmax}_k(z_{k,u,v}), \quad \hat{D}_{u,v} = \sum_{k=1}^{K_D} p_{k,u,v} d_k, \quad (13)$$

where $K_D = 256$ and d_k are metric depth-bin centers. Thus the depth output is represented consistently as a categorical depth probability tensor and its dense metric estimate. In the monocular heads the logits z are produced by a linear probe or DPT-style decoder; in the pose-aware MVS heads they are produced by a regularized projective cost volume before the same softmax-based depth-estimation conversion.

All variants in Table 1 are supervised with the OccuFly target-frame metric depth map D_t and valid-pixel mask Ω_D using the scale-invariant loss in Eq. (25); for Pose-MVS-FiLM, the same supervision is applied to the target-frame MVS estimate $\hat{D}_t^{MVS}$. The table summarizes the feature flow, representative tensor shapes, and learnable components; light-pink boxes mark the parts updated during training, while unmarked backbone features remain frozen.

Table 1: Simplified component flow for the depth-head variants. Light-pink boxes denote the learnable components updated during training; unmarked V-JEPA 2.1 feature extraction remains frozen.

Variant	Data flow with representative shapes
RGB Probe	Linear $I_t \rightarrow F_t^{(47)} [1664, 24, 24] \rightarrow \text{BN} + 1 \times 1 \rightarrow Z_t [256, 24, 24] \rightarrow (\hat{D}_t, P_t^D)$.
Video Probe	Linear $X_t \rightarrow G_t^{(47)} [1664, T', 24, 24] \rightarrow \tilde{F}_t [(1664T'), 24, 24] \rightarrow \text{BN} + 1 \times 1 \rightarrow Z_t [256, 24, 24] \rightarrow (\hat{D}_t, P_t^D)$.
SFP-DPT	$I_t \rightarrow F_t^{(47)} [1664, 24, 24] \rightarrow \text{SFP} \{S_4, S_2, S_1, S_{1/2}\} \rightarrow \text{DPT} \rightarrow (\hat{D}_t, P_t^D)$, $S. \in \{[256, 96, 96], [256, 48, 48], [256, 24, 24], [256, 12, 12]\}$.
DPT	$I_t \rightarrow \{F_t^{(11)}, F_t^{(23)}, F_t^{(37)}, F_t^{(47)}\} \rightarrow \text{DPT reassembly/fusion} \rightarrow (\hat{D}_t, P_t^D)$.
FiLM DPT	$I_t, K_t, h_t \rightarrow \{R_t\}, e_t \rightarrow \text{FiLM}(\{R_t\}, e_t) \rightarrow \text{DPT fusion} \rightarrow (\hat{D}_t, P_t^D)$.
Video DPT	FiLM $X_t, K_t, h_t \rightarrow \{G_t^{(l)} [1664, T', 24, 24]\} \rightarrow \text{DTA} \rightarrow \{\tilde{F}_t^{(l)} [1664, 24, 24]\}, e_t \rightarrow \text{FiLM DPT} \rightarrow (\hat{D}_t, P_t^D)$.
Pose-MVS-FiLM	$\mathcal{V}_t, h_t \rightarrow G_i^{(l)} [1664, T', 24, 24] \rightarrow \text{DTA} \rightarrow \text{pose cost volume} \rightarrow \text{FiLM logits} \rightarrow (\hat{D}_t^{\text{MVS}}, P_t^{D, \text{MVS}})$.

Linear Probe The linear probe is the simplest transfer test in the depth study. It keeps the V-JEPA 2.1 backbone frozen and uses only the learnable head marked in Table 1, following standard frozen-feature dense probing practice and the DINOv3 open-source dense-depth linear head used as implementation reference for the RGB case [21, 29]. Its purpose is diagnostic: if a linear head can recover metric depth, then the frozen representation already exposes monocular depth estimation (MDE) cues in a directly transferable form.

The same diagnostic is applied in RGB and video mode, as summarized in Table 1. In RGB mode, the probe is applied to the final image feature map $F_t^{(47)}$. In video mode, the final video tensor $G_t^{(47)}$ is converted to the video linear-probe feature $\tilde{F}_t$ by folding the temporal tubelet axis into the channel axis before the same type of location-wise head is applied. This video probe therefore tests whether temporal features from V-JEPA 2.1 contain depth cues that are accessible without a nonlinear temporal decoder.

DPT Depth Decoders DPT-style decoders are the structured alternatives to the linear probes. They follow the Dense Prediction Transformer principle of reassembling transformer features into image-like maps and fusing them with convolutional refinement for dense prediction [27]. Table 1 gives the variant-level

data flow, representative shapes, and learnable blocks; the equations below define only the shared DPT intermediates used by the SFP and FiLM variants.

For the DPT path, selected V-JEPA 2.1 layers $\mathcal{L}_G = \{11, 23, 37, 47\}$ are reassembled and projected to a common decoder width $C_D = 256$:

$$R_l = \rho_l \left(F_t^{(l)} \right), \quad R_l \in \mathbb{R}^{C_D \times H_l \times W_l}, \quad l \in \mathcal{L}_G, \quad (14)$$

where ρ_l denotes the DPT reassembly and projection block. These maps are either fused directly by the DPT decoder or passed through FiLM conditioning before fusion.

SFP-DPT. The Simple Feature Pyramid (SFP) variant tests the final-layer-feature hypothesis for V-JEPA 2.1 transfer. V-JEPA 2.1 reports strong dense features with deep supervision [23], so a plausible question is whether the final ViT-G map already carries enough earlier-layer information for depth once it is converted into a multiscale pyramid. To test this, the SFP-DPT branch uses only the final feature map $F_t^{(47)}$ and constructs the multiscale maps listed in Table 1. The pyramid design is inspired by the Simple Feature Pyramid used with plain ViT backbones in ViTDet [14]; we cite it as architectural inspiration rather than as an external implementation dependency.

$$\text{SFP}(F_t^{(47)}) = \{S_4, S_2, S_1, S_{1/2}\}, \quad S_s \in \mathbb{R}^{256 \times (24s) \times (24s)}, \quad s \in \{4, 2, 1, 1/2\}. \quad (15)$$

The resulting pyramid is decoded by the DPT-style depth head marked in Table 1 into 256 depth-bin logits. This variant separates the value of explicit multiscale decoding from the value of using multiple intermediate V-JEPA 2.1 layers.

FiLM-Conditioned DPT for Aerial Depth. The FiLM-conditioned DPT variants test whether altitude and camera calibration should modulate the depth decoder in aerial scenes. This is relevant because UAV altitude and intrinsics affect the mapping from pixels to metric rays, while the frozen V-JEPA 2.1 representation itself is not updated. The flow and tensor shapes for RGB and video FiLM-DPT variants are listed in Table 1; here we define the metadata conditioning applied to the DPT features.

The conditioning input is the normalized metadata vector

$$m_t = \left[\frac{h_t - 40}{10}, \frac{f_x}{W}, \frac{f_y}{H}, \frac{c_x}{W} - 0.5, \frac{c_y}{H} - 0.5 \right] \in \mathbb{R}^5, \quad (16)$$

where h_t is UAV altitude and (f_x, f_y, c_x, c_y) are camera intrinsics expressed relative to the image size used by the branch. A small MLP maps m_t to an embedding $e_t \in \mathbb{R}^{128}$. For each DPT level, learned affine projections produce a scale and shift that modulate the reassembled feature map:

$$\tilde{R}_l = R_l \odot (1 + \gamma_l(e_t)) + \beta_l(e_t), \quad (17)$$

where $\gamma_l(e_t)$ and $\beta_l(e_t)$ have the channel dimension of R_l and are broadcast over spatial locations. Thus FiLM does not change the spatial resolution listed in

Table 1; it applies a learned, metadata-dependent channel-wise scale and shift before DPT fusion. The FiLM projections are initialized so that $\gamma_l \approx 0$ and $\beta_l \approx 0$ at the start of training, making Eq. (17) initially reduce to the unconditioned DPT path. The FiLM-modulated maps $\{\tilde{R}_l\}$ are then fused by DPT refinement blocks and decoded into 256 depth-bin logits.

For video FiLM-DPT, the DPT inputs are first converted from tubelet-aware video features to target-frame feature maps using Dense Temporal Attention (DTA), as shown in Table 1. At each selected layer and spatial site, DTA attends over the temporal tubelet descriptors and returns a target-frame map:

$$\bar{F}_t^{(l)}(:, u, v) = \mu_t^{(l)}(:, u, v) + \alpha \left(\text{Expand}(\text{MLP}(\text{MHA}(q, K, V))) - \mu_t^{(l)}(:, u, v) \right), \quad (18)$$

where $\mu_t^{(l)} = \text{mean}_\tau G_t^{(l)}(:, \tau, :, :)$ and $\alpha = \sigma(a)$ is initialized to approximately 0.1. The resulting maps $\{\bar{F}_t^{(l)}\}_{l \in \mathcal{L}_G}$ are passed to the same FiLM-conditioned DPT decoder as the RGB variant.

Pose-MVS-FiLM Pose-MVS-FiLM is the pose-aware multiview depth branch in Table 1. It keeps the V-JEPA 2.1 backbone frozen and uses calibrated target/source views to construct a depth-indexed cost volume, following the reference-view matching principle used in learned MVS cost-volume methods such as MVS-Net [35]. Let $\mathcal{V}_t = \{(I_i, K_i, T_{t \rightarrow i})\}_{i \in \mathcal{S}_t}$ denote the calibrated source views for target frame t . For target pixel $\mathbf{u} = (u, v, 1)^\top$ and metric depth-bin center d_k , the target-frame 3D point and source-view projection are

$$\mathbf{x}_{t,k,u,v} = d_k K_t^{-1} \mathbf{u}, \quad \mathbf{q}_{i,k,u,v} = \pi \left(K_i T_{t \rightarrow i} \begin{bmatrix} \mathbf{x}_{t,k,u,v} \\ 1 \end{bmatrix} \right), \quad (19)$$

where $\pi(\cdot)$ is perspective projection. Let A_t and A_i denote target/source feature maps after the video-feature adaptation shown in Table 1. Source features are sampled at the projected locations, compared with the target feature, and aggregated over source views into a depth-indexed cost volume,

$$\tilde{A}_{i,k,u,v} = \text{sample}(A_i, \mathbf{q}_{i,k,u,v}), \quad C_{t,k,u,v} = \text{Agg}_{i \in \mathcal{S}_t} \text{corr} \left(A_t(:, u, v), \tilde{A}_{i,k,u,v} \right). \quad (20)$$

The cost volume is regularized by a 3D CNN and mapped to FiLM-conditioned target-frame depth logits,

$$Z_t^{\text{MVS}} = \phi_{\text{logit}}(\psi_{3D}(C_t), e_t),$$

$$P_t^{D, \text{MVS}} = \text{softmax}_k Z_t^{\text{MVS}}, \quad \hat{D}_{t,u,v}^{\text{MVS}} = \sum_{k=1}^{K_D} P_{t,k,u,v}^{D, \text{MVS}} d_k. \quad (21)$$

This branch uses the same $K_D = 256$ metric depth-bin centers and target-frame depth supervision as Eq. (13). CVP-MVSNet and CasMVSNet are treated as related cascaded MVS literature where cited elsewhere; the implementation

Table 2: Depth-estimation variants and the transfer question isolated by each design. Learnable components are shown in Table 1.

Variant	Purpose
Linear probe	Tests whether metric aerial depth is linearly accessible from frozen V-JEPA 2.1 representations.
SFP-DPT	Tests whether the final V-JEPA 2.1 map can be converted into an effective dense pyramid without using intermediate layers.
DPT	Tests whether selected intermediate layers improve dense metric depth over final-layer probing.
FiLM-DPT	Tests whether altitude/intrinsics conditioning improves RGB or video depth decoding.
Pose-MVS-FiLM	Tests whether calibrated multiview matching and meta-data conditioning improve the geometric prior.

described here is a single cost-volume branch rather than a cascade of those methods.

Table 2 summarizes the depth variants and the question each variant is designed to answer. The learnable components for these variants are marked in Table 1, so this table focuses only on experimental motivation. The experiments form a controlled sequence from minimal representation probing to structured spatio-temporal decoding.

Altitude-Interpolation Depth Evaluation The standard OccuFly dataset split uses scenes 01–05 for training, scenes 06–07 for validation, and scenes 08–09 for testing, with altitudes 30 m, 40 m, and 50 m. We also include an altitude-interpolation protocol in which training uses 30 m and 50 m samples from scenes 01–08, validation uses 40 m samples from scenes 01–04, and testing uses 40 m samples from scenes 05–08:

$$\mathcal{D}_{\text{train}} = \{s_t : h_t \in \{30, 50\}, \text{scene}(t) \in \{01, \dots, 08\}\}, \quad (22)$$

$$\mathcal{D}_{\text{val}} \subset \{s_t : h_t = 40, \text{scene}(t) \in \{01, 02, 03, 04\}\}, \quad (23)$$

$$\mathcal{D}_{\text{test}} \subset \{s_t : h_t = 40, \text{scene}(t) \in \{05, 06, 07, 08\}\}. \quad (24)$$

This split is a scale-generalization stress test: it reports model depth prediction errors at the unseen middle altitude. It is especially relevant for FiLM-conditioned depth heads, because their metadata pathway is intended to use UAV altitude at inference time; performance at 40 m after training on 30 m and 50 m tests whether that conditioning supports interpolation across altitude rather than only fitting observed flight heights.

Losses The resulting depth map is resized to the supervision resolution and trained with a valid-pixel scale-invariant logarithmic loss. Let Ω_D be the set of

valid supervised pixels after resizing, $N = |\Omega_D|$, and $i \in \Omega_D$ index one valid pixel. For predicted depth $\hat{D}_i$, ground-truth OccuFly depth D_i , and log-depth residual $g_i = \log \hat{D}_i - \log D_i$, the loss is

$$\mathcal{L}_{\text{sig}} = \sqrt{\frac{1}{N} \sum_{i \in \Omega_D} g_i^2 - \lambda_{\text{si}} \left(\frac{1}{N} \sum_{i \in \Omega_D} g_i \right)^2} + \epsilon, \quad (25)$$

where λ_{si} controls the scale-invariant correction term and ϵ is a small positive constant for numerical stability. The term inside the square root is the scale-invariant log-depth objective of Eigen et al. [7] when $\lambda_{\text{si}} = 0.5$; the square root and ϵ give the numerically stabilized SiLog-style form used here. We use this loss because aerial metric depth contains unavoidable global-scale ambiguity, while the altitude/intrinsics conditioning experiments test whether that ambiguity can be reduced by physically meaningful metadata.

Evaluation Metrics Depth models are evaluated only over valid pixels. The manuscript uses root mean squared error (RMSE), absolute relative error (AbsRel), and log-RMSE:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i \in \Omega_D} (\hat{D}_i - D_i)^2}, \quad \text{AbsRel} = \frac{1}{N} \sum_{i \in \Omega_D} \frac{|\hat{D}_i - D_i|}{D_i}, \quad (26)$$

$$\text{LogRMSE} = \sqrt{\frac{1}{N} \sum_{i \in \Omega_D} (\log \hat{D}_i - \log D_i)^2}. \quad (27)$$

For video-depth runs, the evaluation script also supports MAE, squared-relative error, and threshold accuracies $\delta_1, \delta_2, \delta_3$ when enabled.

3.4 Semantic Scene Completion

The SSC benchmark isolates two factors that are otherwise coupled in end-to-end completion systems. The image-context pathway contributes semantic visual features, while the geometry pathway contributes monocular depth, depth distributions, or pose-aware MVS priors. Both pathways are then lifted into a common 3D representation before fusion and semantic occupancy prediction.

Semantic scene completion (SSC) is evaluated after depth estimation because the depth branches provide geometry priors for downstream 3D semantic occupancy. The SSC experiments test whether those priors, together with the corresponding image-context or F2D features, transfer into a 3D completion task. The purpose is not to introduce a completely new SSC architecture. Instead, we use published SSC designs whose image-context and geometry pathways are exposed enough to support controlled feature-transfer studies.

CGFormer and FoundationSSC are selected consciously even though their published forms use stereo-derived geometry priors rather than the monocular MDE or pose-aware MVS depth estimators targeted in this chapter. The reason is methodological: both expose a useful separation between image-context information and geometry information. CGFormer combines an image encoder, depth-context components including depth-aware view transformation, CGVT, a 3D local-global encoder, and a decoding head [37]. FoundationSSC is a dual-decoupling framework with a foundation encoder, decoupled semantic-context and geometric pathways, hybrid view transformation, axis-aware fusion, and a decoding head [6]. We adapt these published designs for controlled evaluation.

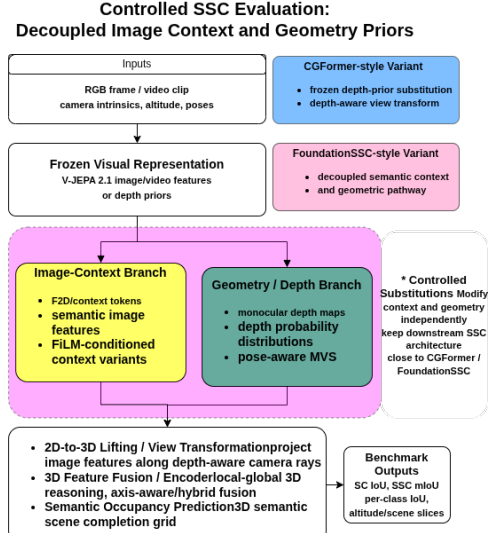


Fig. 3: Conceptual SSC flow with independently varied image-context and geometry-depth pathways.

CGFormer Variant: Frozen Depth Prior Substitution The modification is to use precomputed depth maps from frozen pretrained depth heads as replacement geometry priors for CGFormer’s original stereo-depth or depth-map input slot. The depth-prior sources considered are from Depth Anything V2 fine tuned on Occufly (cite occufly dataset) and V-JEPA 2.1 depth head DPT-FiLM trained on Occufly. SSC metrics like occupied mean intersection over union (mIoU) for

Table 3: SSC variants and the geometric prior isolated by each design.

Variant	Geometric prior	Question isolated
CGFormer compar- ison	Exported monocular pri- ors	Do frozen V-JEPA 2.1-derived depth priors change a fixed CGFormer-style SSC evaluation path?

measure of sematic class matching and intersection over union (IoU) for measure
of occupied occupany are reported as a preliminary comparison.

FoundationSSC Variant FoundationSSC is used as a stronger published
follow-up baseline because it explicitly decouples semantic-context and geometric
information at both the source and pathway levels [6]. In its published imple-
mentation, a frozen FoundationStereo encoder provides image-context features,
cost-volume geometry features, and a dense depth map. Compared with CG-
Former, FoundationSSC is reported in the published comparison to improve on
SemanticKITTI and SSCBench-KITTI-360, with the gains attributed to the de-
coupled foundation encoder-pathways, learned disparity-to-depth volume map-
ping, hybrid view transformation, and axis-aware fusion [6,37]. We therefore use
it as a suitable architecture for ablation studies.

Considered Losses

Evaluation Metrics SSC evaluation uses IoU and mIoU, following the pub-
lished CGFormer and FoundationSSC evaluation protocol [6,37]. IoU is the ge-
ometric occupied-voxel completion metric: all non-empty labels are treated as
occupied, and the score measures occupied-vs-empty completion. mIoU is the
voxel-wise semantic completion metric averaged over semantic classes.

4 Experimental Setup and Results

4.1 Dataset and Tasks

We evaluate on OccuFly [8] for monocular metric depth estimation and seman-
tic scene completion (SSC). For the standard protocol, we use scenes 01–05 for
training, scenes 06–07 for validation, and scenes 08–09 for testing, across alti-
tudes of 30, 40, and 50 m. The standard depth split contains 14,804 training
frames, 1,965 validation frames, and 3,842 test frames. RGB inputs are resized
to 384×384 . Depth values are represented in meters, clipped to $[0.001, 80]$, and

Table 4: OccuFly test monocular depth estimation comparison. Published baseline values for MapAnythingV1.1, Metric3Dv2, Depth Anything 2, and Depth Anything 3 follow Table 5 of OccuFly [8]. Light pink marks the best RMSE.

Method	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	SILog $\downarrow$
Linear Probe RGB	0.651	0.958	0.996	0.189	7.808	6.457	0.226
Linear Probe Video 4 frame clip	0.692	0.953	0.994	0.190	7.226	5.996	0.220
SFP_FiLM_DPT	0.808	0.953	0.993	0.143	5.160	3.732	0.189
DPT	0.469	0.795	0.898	0.267	11.471	9.227	0.411
FiLM_DPT_RGB	0.833	0.971	0.998	0.129	4.621	3.523	0.164
DTA_FiLM_DPT_Video	0.805	0.970	0.998	0.136	4.878	3.635	0.167
DepthAnything2_OccuFly [8]	0.834	0.976	0.999	0.134	4.844	4.193	0.112
Metric3Dv2_OccuFly [8]	0.278	0.806	0.985	0.381	13.643	13.134	0.098
MapAnythingV1.1 [13]	0.000	0.000	0.003	0.799	30.068	29.309	0.069
Metric3Dv2 [10]	0.073	0.208	0.455	0.471	19.578	18.409	0.156
DepthAnything2 [34]	0.002	0.015	0.059	0.729	28.382	27.392	0.192
DepthAnything3 [17]	0.000	0.005	0.141	0.591	22.615	22.019	0.043

Table 5: Best depth result for the altitude-interpolation split: training uses all scenes 01–09 at 30 m and 50 m altitude, validation uses scenes 01–04 at 40 m altitude, and testing uses scenes 05–08 at 40 m altitude.

Backbone & Depth head	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	SILog $\downarrow$
V-JEPA 2.1 RGB FiLM_DPT_RGB	0.9653	0.9997	0.9999	0.1021	4.7071	3.9570	0.119599

evaluated only over finite positive pixels that remain valid after mask-aware resizing.

For SSC, OccuFly labels are defined on a voxel grid of shape $192 \times 128 \times 128$ in WHD order. The grid spans $x \in [-48, 48]$ m, $y \in [-32, 32]$ m, and $z \in [0, 64]$ m, corresponding to a voxel size of 0.5 m.

4.2 MDE benchmarking results

All experiments are run on an NVIDIA DGX node with 8 NVIDIA H100 80GB GPUs.

4.3 SSC benchmarking results

Table 8 reports semantic scene completion performance on OccuFly. The published comparison rows reproduce the altitude-wise and overall values from OccuFly Table 4 [8] for Symphonies [12] and DISC [19]. The CGFormer and FoundationSSC rows report values exported by the local SSC benchmark dashboard. Fig. 4 further breaks down the class-wise IoU for the same evaluated SSC variants, making visible which semantic categories drive the aggregate SSC mIoU differences. Fig. 6 shows a representative qualitative comparison of the corresponding depth and semantic occupancy outputs.

Table 6: OccuFly test MDE comparison grouped by altitude, with Metric3Dv2-OccuFly and DepthAnything2-OccuFly altitude-wise values appended from OccuFly Table 12 [8]. Light pink marks the best RMSE within each altitude group.

Group Method		$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	SILog $\downarrow$
30	Linear Probe RGB	0.743	0.952	0.986	0.205	6.280	5.391	17.585
	Linear Probe Video 4 frame clip	0.512	0.933	0.979	0.266	7.942	7.002	17.676
	SFP_FiLM_DPT	0.889	0.961	0.991	0.111	4.022	2.847	16.638
	DPT	0.415	0.889	0.940	0.295	9.505	8.238	35.043
	FiLM_DPT_RGB	0.917	0.980	0.999	0.097	3.448	2.571	13.185
	DTA_FiLM_DPT_Video	0.944	0.982	0.999	0.081	2.939	2.051	12.356
	Metric3Dv2-OccuFly [8]	0.460	0.757	0.992	0.347	10.917	10.215	0.101
	DepthAnything2-OccuFly [8]	0.919	0.982	0.998	0.103	3.108	2.666	0.086
40	Linear Probe RGB	0.730	0.943	0.998	0.180	5.851	4.804	21.671
	Linear Probe Video 4 frame clip	0.720	0.924	0.996	0.185	5.763	4.510	20.695
	SFP_FiLM_DPT	0.729	0.911	0.988	0.186	5.665	4.056	21.991
	DPT	0.577	0.794	0.905	0.239	9.106	6.916	39.112
	FiLM_DPT_RGB	0.761	0.945	0.995	0.158	4.804	3.588	18.577
	DTA_FiLM_DPT_Video	0.737	0.938	0.997	0.173	5.191	3.800	19.481
	Metric3Dv2-OccuFly [8]	0.209	0.802	0.981	0.395	13.138	12.625	0.096
	DepthAnything2-OccuFly [8]	0.795	0.957	0.998	0.148	4.486	3.823	0.119
50	Linear Probe RGB	0.524	0.978	1.000	0.192	9.981	8.728	17.395
	Linear Probe Video 4 frame clip	0.741	0.994	1.000	0.161	8.244	7.158	16.681
	SFP_FiLM_DPT	0.858	0.995	1.000	0.110	5.038	3.777	13.753
	DPT	0.375	0.754	0.872	0.285	14.243	12.179	37.819
	FiLM_DPT_RGB	0.873	0.995	1.000	0.113	4.872	3.879	12.947
	DTA_FiLM_DPT_Video	0.817	0.998	1.000	0.122	5.213	4.165	13.538
	Metric3Dv2-OccuFly [8]	0.283	0.831	0.987	0.379	15.373	14.950	0.099
	DepthAnything2-OccuFly [8]	0.844	0.997	1.000	0.129	5.985	5.261	0.114

5 Discussion and Conclusion

References

1. Assran, M., Duval, Q., Misra, I., Bojanowski, P., Vincent, P., Rabbat, M., LeCun, Y., Ballas, N.: Self-supervised learning from images with a joint-embedding predictive architecture. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (2023) 6

2. Assran, M., Bardes, A., Fan, D., Garrido, Q., Howes, R., Komeili, M., Muckley, M., Rizvi, A., Roberts, C., Sinha, K., et al.: V-jepa 2: Self-supervised video models enable understanding, prediction and planning. arXiv preprint arXiv:2506.09985 (2025) 3, 6, 11

3. Bardes, A., Garrido, Q., Ponce, J., Chen, X., Rabbat, M., LeCun, Y., Assran, M., Ballas, N.: Revisiting feature prediction for learning visual representations from video. arXiv preprint arXiv:2404.08471 (2024) 3, 6

4. Bhat, S.F., Birkel, R., Wofk, D., Wonka, P., Müller, M.: Zoedepth: Zero-shot transfer by combining relative and metric depth. arXiv preprint arXiv:2302.12288 (2023) 5

Table 7: OccuFly test MDE comparison grouped by scene.

Group	Method	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	SILog $\downarrow$
scene_08	Linear Probe RGB	0.752	0.985	0.999	0.159	6.972	5.715	18.664
	Linear Probe Video 4 frame clip	0.665	0.983	0.999	0.183	7.465	6.358	19.247
	SFP_FiLM_DPT	0.976	0.998	1.000	0.069	3.014	2.468	7.813
	DPT	0.601	0.946	0.976	0.212	8.949	7.353	27.068
	FiLM_DPT_RGB	0.975	0.997	1.000	0.067	2.934	2.392	7.869
	DTA_FiLM_DPT_Video	0.982	0.998	1.000	0.055	2.566	2.010	7.090
scene_09	Linear Probe RGB	0.610	0.947	0.995	0.201	8.121	6.757	23.551
	Linear Probe Video 4 frame clip	0.702	0.941	0.993	0.192	7.129	5.851	22.872
	SFP_FiLM_DPT	0.741	0.935	0.990	0.173	5.805	4.242	20.643
	DPT	0.415	0.734	0.866	0.289	12.343	9.983	42.988
	FiLM_DPT_RGB	0.776	0.960	0.997	0.155	5.147	3.979	17.416
	DTA_FiLM_DPT_Video	0.734	0.958	0.998	0.169	5.542	4.287	17.872

5. Cao, A.Q., de Charette, R.: Monoscene: Monocular 3d semantic scene completion. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (2022) 2, 6

6. Chen, S., Sui, W., Zhang, B., Boukhers, Z., See, J., Yang, C.: Unleashing semantic and geometric priors for 3d scene completion. In: Proceedings of the AAAI Conference on Artificial Intelligence (2026) 2, 3, 7, 18, 19

7. Eigen, D., Puhrsch, C., Fergus, R.: Depth map prediction from a single image using a multi-scale deep network. In: Advances in Neural Information Processing Systems (2014) 5, 17

8. Gross, M., Matha, S.B., Fahmy, A., Song, R., Cremers, D., Meess, H.: OccuFly: A 3d vision benchmark for semantic scene completion from the aerial perspective. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (2026) 2, 4, 7, 8, 19, 20, 21, 26

9. He, K., Chen, X., Xie, S., Li, Y., Dollár, P., Girshick, R.: Masked autoencoders are scalable vision learners. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (2022) 6

10. Hu, M., Yin, W., Zhang, C., Cai, Z., Long, X., Chen, H., Wang, K., Yu, G., Shen, C., Shen, S.: Metric3D v2: A versatile monocular geometric foundation model for zero-shot metric depth and surface normal estimation. IEEE Transactions on Pattern Analysis and Machine Intelligence 46(12), 10579–10596 (2024) 20

11. Huang, Y., Zheng, W., Zhang, Y., Zhou, J., Lu, J.: Tri-perspective view for vision-based 3d semantic occupancy prediction. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (2023) 2, 7

12. Jiang, H., Cheng, T., Gao, N., Zhang, H., Lin, T., Liu, W., Wang, X.: Symphonize 3d semantic scene completion with contextual instance queries. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2024) 20, 26

13. Keetha, N., Müller, N., Schönberger, J., Porzi, L., Zhang, Y., Fischer, T., Knapitsch, A., Zauss, D., Weber, E., Antunes, N., Luiten, J., Lopez-Antequera, M., Rota Bulò, S., Richardt, C., Ramanan, D., Scherer, S., Kotschieder, P.: MapAnything: Universal feedforward metric 3d reconstruction. In: International Conference on 3D Vision (3DV) (2026) 20

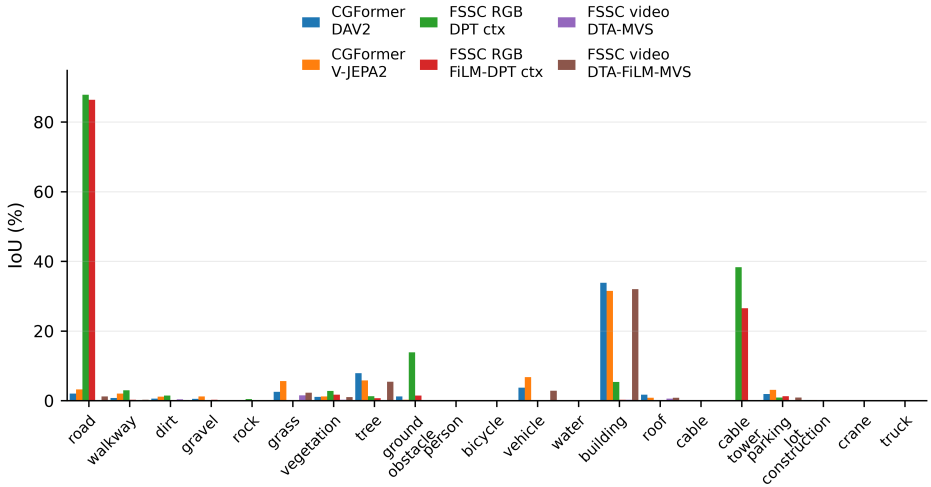


Fig. 4: OccuFly SSC per-class IoU bar chart for the evaluated CGFormer and FoundationSSC variants.

14. Li, Y., Mao, H., Girshick, R., He, K.: Exploring plain vision transformer backbones for object detection. In: European Conference on Computer Vision (2022) 3, 5, 14
15. Li, Y., Yu, Z., Choy, C., Xiao, C., Alvarez, J.M., Fidler, S., Feng, C., Anandkumar, A.: Voxformer: Sparse voxel transformer for camera-based 3d semantic scene completion. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (2023) 2, 7
16. Li, Y., Ge, Z., Yu, G., Yang, J., Wang, Z., Shi, Y., Sun, J., Li, Z.: BEVDepth: Acquisition of reliable depth for multi-view 3d object detection. In: Proceedings of the AAAI Conference on Artificial Intelligence (2023) 7
17. Lin, H., Chen, S., Liew, J.H., Chen, D.Y., Li, Z., Shi, G., Feng, J., Kang, B.: Depth anything 3: Recovering the visual space from any views. arXiv preprint arXiv:2511.10647 (2025) 20
18. Lin, L., Liu, Y., Hu, Y., Yan, X., Xie, K., Huang, H.: Capturing, reconstructing, and simulating: The urbanscene3d dataset. In: European Conference on Computer Vision (2022) 2, 4
19. Liu, E., Yu, E., Chen, S., Tao, W.: Disentangling instance and scene contexts for 3d semantic scene completion (2025) 20, 26
20. Lyu, Y., Vosselman, G., Xia, G.S., Yilmaz, A., Yang, M.Y.: UAVid: A semantic segmentation dataset for uav imagery. ISPRS Journal of Photogrammetry and Remote Sensing (2020) 2, 4
21. Meta AI Research: DINOv3 dense depth linearhead implementation. https://github.com/facebookresearch/dinov3/blob/main/dinov3/eval/depth/models/linear_head.py (2025), open-source frozen-feature dense linear head using batch normalization and a 1×1 convolution 10, 13
22. Meta AI Research: facebookresearch/vjepa2: Pytorch code and models for v-jepa 2 and v-jepa 2.1. <https://github.com/facebookresearch/vjepa2> (2025), official implementation, including attentive probe code 11

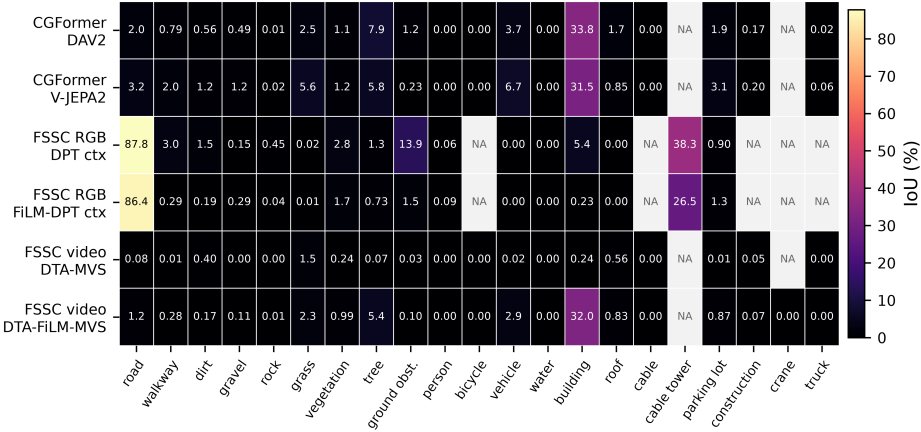


Fig. 5: OccuFly SSC per-class IoU heatmap for the evaluated CGFormer and FoundationSSC variants.

23. Mur-Labadia, L., Muckley, M., Bar, A., Assran, M., Sinha, K., Rabbat, M., LeCun, Y., Ballas, N., Bardes, A.: V-jepa 2.1: Unlocking dense features in video self-supervised learning. arXiv preprint arXiv:2603.14482 (2026) [2](#), [6](#), [10](#), [11](#), [14](#)

24. Oquab, M., Darcet, T., Moutakanni, T., Vo, H.V., Szafraniec, M., Khalidov, V., Fernandez, P., Haziza, D., Massa, F., El-Nouby, A., et al.: DINOv2: Learning robust visual features without supervision. Transactions on Machine Learning Research (2024) [6](#)

25. Perez, E., Strub, F., de Vries, H., Dumoulin, V., Courville, A.: FiLM: Visual reasoning with a general conditioning layer. In: Proceedings of the AAAI Conference on Artificial Intelligence (2018) [3](#), [5](#)

26. Phillion, J., Fidler, S.: Lift, splat, shoot: Encoding images from arbitrary camera rigs by implicitly unprojecting to 3d. arXiv preprint arXiv:2008.05711 (2020) [3](#), [7](#)

27. Ranftl, R., Bochkovskiy, A., Koltun, V.: Vision transformers for dense prediction. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (2021) [3](#), [5](#), [10](#), [13](#)

28. Ranftl, R., Lasinger, K., Hafner, D., Schindler, K., Koltun, V.: Towards robust monocular depth estimation: Mixing datasets for zero-shot cross-dataset transfer. IEEE Transactions on Pattern Analysis and Machine Intelligence (2020) [5](#)

29. Siméoni, O., Vo, H.V., Seitzer, M., Baldassarre, F., Oquab, M., Jose, C., Khalidov, V., Szafraniec, M., Yi, S., Ramamonjisoa, M., et al.: DINOv3. arXiv preprint arXiv:2508.10104 (2025) [6](#), [10](#), [13](#)

30. Song, S., Yu, F., Zeng, A., Chang, A.X., Savva, M., Funkhouser, T.: Semantic scene completion from a single depth image. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (2017) [2](#), [6](#)

31. Wang, X., Zhu, Z., Xu, W., Zhang, Y., Wei, Y., Chi, X., Ye, Y., Du, D., Lu, J., Wang, X.: Openoccupancy: A large scale benchmark for surrounding semantic occupancy perception. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (2023) [2](#), [7](#)

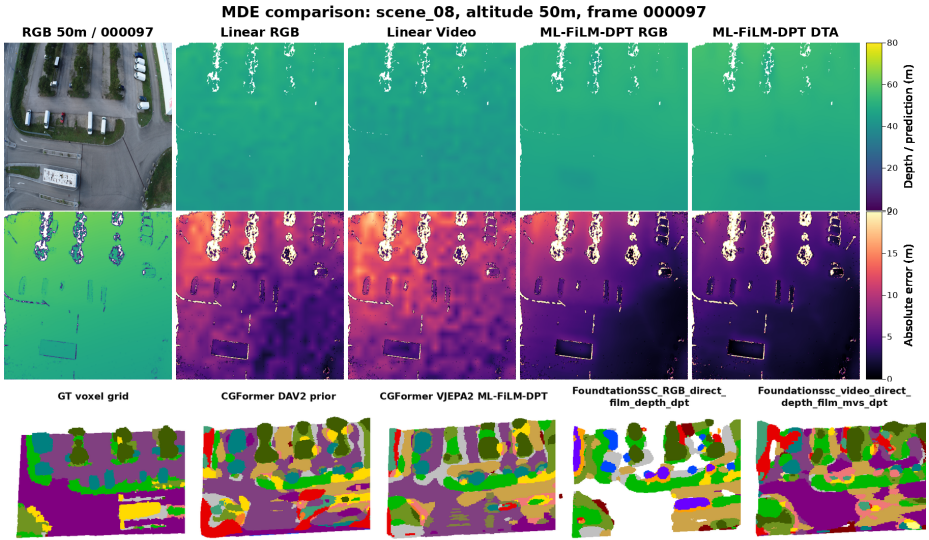


Fig. 6: Qualitative OccuFly example for scene 08 at 50 m altitude, frame 000097, showing the depth-estimation context and semantic occupancy comparison used to inspect SSC failure modes beyond aggregate IoU and mIoU.

32. Wei, Y., Zhao, L., Zheng, W., Zhu, Z., Zhou, J., Lu, J.: Surroundocc: Multi-camera 3d occupancy prediction for autonomous driving. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (2023) 2, 7
33. Yang, L., Kang, B., Huang, Z., Xu, X., Feng, J., Zhao, H.: Depth anything: Unleashing the power of large-scale unlabeled data. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (2024) 5
34. Yang, L., Kang, B., Huang, Z., Zhao, Z., Xu, X., Feng, J., Zhao, H.: Depth anything v2. In: Advances in Neural Information Processing Systems (2024) 5, 7, 20
35. Yao, Y., Luo, Z., Li, S., Fang, T., Quan, L.: MVSNet: Depth inference for unstructured multi-view stereo. In: Proceedings of the European Conference on Computer Vision (ECCV). pp. 767–783 (2018) 3, 5, 7, 15
36. Yao, Y., Luo, Z., Li, S., Shen, T., Fang, T., Quan, L.: Recurrent MVSNet for high-resolution multi-view stereo depth inference. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 5525–5534 (2019) 5, 7
37. Yu, Z., Zhang, R., Ying, J., Yu, J., Hu, X., Luo, L., Cao, S., Shen, H.: Context and geometry aware voxel transformer for semantic scene completion. In: Advances in Neural Information Processing Systems (2024) 2, 3, 7, 18, 19

Table 8: OccuFly SSC benchmark comparison by altitude. Published Symphonies and DISC values follow OccuFly Table 4 [8]; CGFormer and FoundationSSC values are from the local SSC benchmark report. For our variants, light beige marks the best SC IoU and light pink marks the best SSC mIoU within each altitude group and the overall All group.

Altitude Method		SC IoU $\uparrow$	SSC mIoU $\uparrow$
<i>Published OccuFly Table 4 baselines</i>			
50 m	Symphonies [12]	15.88	0.58
50 m	DISC [19]	31.10	2.20
40 m	Symphonies [12]	10.71	0.52
40 m	DISC [19]	27.85	1.77
30 m	Symphonies [12]	13.22	0.76
30 m	DISC [19]	26.88	2.23
All	Symphonies [12]	13.68	0.58
All	DISC [19]	29.52	2.04
<i>Our CGFormer and FoundationSSC variants</i>			
30 m	CGFormer DAV2 prior	36.10	4.22
30 m	CGFormer V-JEPA 2.1 FiLM_DPT_RGB prior	36.70	4.39
30 m	FSSC_RGB_context_DPT_depth_Film_DPT	33.21	5.16
30 m	FSSC_RGB_context_FiLM_DPT_depth_FiLM_DPT	6.74	0.26
30 m	FSSC_video_context_DTA_DPT_depth_DTA_mvs	2.84	0.18
30 m	FSSC_video_context_DTA_DPT_depth_DTA_FiLM_mvs	35.59	5.06
40 m	CGFormer DAV2 prior	38.42	2.86
40 m	CGFormer V-JEPA 2.1 FiLM_DPT_RGB prior	29.05	2.12
40 m	FSSC_RGB_context_DPT_depth_Film_DPT	27.85	4.08
40 m	FSSC_RGB_context_FiLM_DPT_depth_FiLM_DPT	22.15	1.69
40 m	FSSC_video_context_DTA_DPT_depth_DTA_mvs	12.30	0.87
40 m	FSSC_video_context_DTA_DPT_depth_DTA_FiLM_mvs	29.22	3.97
50 m	CGFormer DAV2 prior	35.56	2.81
50 m	CGFormer V-JEPA 2.1 FiLM_DPT_RGB prior	42.78	4.06
50 m	FSSC_RGB_context_DPT_depth_Film_DPT	21.16	2.53
50 m	FSSC_RGB_context_FiLM_DPT_depth_FiLM_DPT	19.51	0.75
50 m	FSSC_video_context_DTA_DPT_depth_DTA_mvs	5.21	0.21
50 m	FSSC_video_context_DTA_DPT_depth_DTA_FiLM_mvs	22.28	2.59
All	CGFormer DAV2 prior	36.77	3.05
All	CGFormer V-JEPA 2.1 FiLM_DPT_RGB prior	36.51	3.31
All	FSSC_RGB_context_DPT_depth_Film_DPT	37.68	3.46
All	FSSC_RGB_context_FiLM_DPT_depth_FiLM_DPT	35.47	1.73
All	FSSC_video_context_DTA_DPT_depth_DTA_mvs	24.55	0.17
All	FSSC_video_context_DTA_DPT_depth_DTA_FiLM_mvs	32.55	2.36